MC10

SETUP GUIDE

MC10 Controller

for Fractal Audio FM3 & FM3 MkII

Custom firmware for the PaintAudio MIDI Captain 10. Six dedicated effect switches, eight scenes, full looper control — and a display that always shows what the FM3 is actually doing.

6	8	20
EFFECT SWITCHES	SCENES	MINUTES TO SET UP

No coding needed. Change a few settings on your FM3, load the firmware, then set the pedal up from your browser.

Everything is reversible — the original firmware can be restored at any time. See section 5.

1 Before you start

What you need

- PaintAudio MIDI Captain 10 STD — the ten-switch model
- Fractal FM3 or FM3 MkII
- **Two** 5-pin MIDI cables — pedal MIDI OUT → FM3 MIDI IN, and FM3 MIDI OUT → pedal MIDI IN
- A USB cable that carries data — many are charge-only
- Chrome or Edge on a computer, to set the pedal up (section 6)
- Optional: FM3-Edit — makes the FM3 settings much quicker

> GOOD TO KNOW
>
> The two cables cross over: an OUT always goes into an IN, in both directions. The pedal sends your button presses to the FM3 and listens to what the FM3 answers, and that is what keeps the display in step with the FM3. A cable the wrong way round, or only one of the two, is the single most common setup mistake, and the display will simply show `NOT SYNC`.

2 Settings on the FM3

Everything here lives under *Setup → MIDI/Remote*. You can do it from the front panel, but FM3-Edit is far quicker.

The essentials

SETTING	SET TO	WHY
MIDI Channel	1	The controller talks on channel 1 — this can be changed in the editor
Program Change	On	Lets the controller change presets
Ignore Redundant PC	Off	So repeating a preset still works

Turn these off

These make the FM3 send a constant stream of messages back to the pedal. Nothing breaks if they stay on — the connection is just cleaner without them.

SETTING	SET TO	WHAT IT IS
Send MIDI Clock	Off	Tempo sync for other gear
Send Realtime SysEx	Off	Continuous status messages
Send MIDI PC	Off	Preset changes sent outward

3 Assigning the effects

The controller has six effect switches. You tell the FM3 which block each one turns on and off.

The easy way — MIDI Learn

1. On the FM3, open *Setup* → *MIDI/Remote* → *Bypass*
2. Find the block you want — Drive 1, Delay 1, Reverb 1…
3. Click into its CC field
4. Press the switch on the pedal — the FM3 fills in the number itself

Repeat for each of the six effects. That's all there is to it.

Or type the numbers by hand

BLOCK	CC	WHERE IT APPEARS
Drive 1	37	Switch 1 — short press
Compressor 1	38	Switch 1 — long press
Chorus 1	41	Switch 2 — short press
Flanger 1	42	Switch 2 — long press
Delay 1	40	Switch 3 — short press
Reverb 1	43	Switch 3 — long press

And two more, under the *Channel* tab:

WHAT	CC	USED BY
Amp 1 channel	101	Switch combinations A+B and B+C
Delay 1 channel	102	Switch combinations C+D and D+DOWN

> THESE ARE ONLY DEFAULTS
> Every number here, the MIDI channel, and which FM3 block sits on which switch can be changed in the configuration editor and sent to the pedal in seconds — see section 6. Nothing has to be recompiled.

4 Loading the firmware

Inside the MC10 is a Raspberry Pi Pico. Loading firmware means copying one file across — but first the pedal has to be put into bootloader mode, and that part is done PaintAudio's way.

> FOLLOW THE OFFICIAL DOCUMENT
> PaintAudio publish a step-by-step guide called “Update Process — Please read carefully”. It comes with the firmware download and has screenshots for every step, on Windows and macOS. Use that document for the procedure itself. What follows here is only a summary, so you know what to expect.

Before anything else: back up

> THIS CAN ONLY BE DONE BEFORE YOU SWITCH FIRMWARE
> Hold Key 1 while powering on — the **PaintAudio** firmware opens its file storage as a drive. Copy everything off it into a folder you'll remember. Once this firmware is loaded, Key 1 does something else (it opens config mode), so there is no second chance.

What the official procedure looks like

1. Connect the pedal to the computer and power it on
2. Open `MIDICAPTAINBOOT.HTML` in Chrome — Safari is not supported
3. Click BOOT, pick the serial port that appears (TinyUSB Serial or Pico), and connect
4. A drive named `RPI-RP2` appears — that is bootloader mode
5. Copy the `.uf2` file onto it. On macOS drag it onto the drive icon, otherwise the drive may end up read-only

The drive disappears and the pedal restarts. That's normal, not an error.

> NOTHING TO FORMAT, NOTHING TO COPY
> The official document also describes formatting the pedal's storage and copying a folder of files onto it. That belongs to PaintAudio's own firmware, not to this one. This firmware is a single `.uf2` — everything is inside it. Load the file and you're done, then set it up from the browser (section 6).

> IF YOU'D RATHER OPEN THE PEDAL
> There is a hardware route as well: the Pico's BOOTSEL contacts are reachable from the underside of the board, and shorting them while powering on brings up `RPI-RP2` directly. This needs soldering experience and voids any warranty — the browser method above does the same job without opening anything.

5 Going back

Nothing here is permanent. Whatever was on the pedal can be put back.

Back to the PaintAudio firmware

1. Download the current firmware package for the MIDI Captain 10 STD from PaintAudio
2. Follow the same official update document — identical procedure, only with their `.uf2` instead of this one
3. Once the PaintAudio firmware is running again, hold Key 1 on power-up and copy back the files you saved earlier

Back to CircuitPython

If your pedal originally ran CircuitPython, put it into `RPI-RP2` mode the same way and copy the CircuitPython `.uf2` across. A drive named `CIRCUITPY` appears, and your saved files go back there.

> THE PEDAL CANNOT BE BRICKED
> Bootloader mode lives in the chip itself, not in the firmware. Even with nothing installed at all, the browser tool — or the hardware route — will still bring up `RPI-RP2`.

6 Changing the layout

Open the editor file in **Chrome or Edge on a computer**. Nothing to install, works offline, nothing is uploaded anywhere.

YOU CAN CHANGE	EXAMPLES
Switch assignments	Which effect sits on which switch, short and long press
FM3 blocks	Which block each of the six effect slots follows — Drive, Comp, Pitch, Phaser, Rotary, Synth, Tremolo…
CC numbers	MIDI channel, effect CCs, looper CCs, expression pedals, volume encoders
Channel switching	The two channel pairs: on or off, and which block's channel the display follows
Display	Colours, text sizes, row positions, labels
Startup	Which preset to jump to at power-on — or none at all
Warnings	LED colour and flash count when a block is missing

Putting a config on the pedal

The pedal has two modes. In **config mode** it talks to the editor and you can send, pick and delete configs. In normal mode it is your controller and you can only *choose* between configs that are already on it. Both are described here.



Sending a config — step by step

1. **Connect the pedal to the computer** with a USB cable that carries data. The pedal is switched off for now.
2. **Start it in config mode.** Hold **SWITCH 1** while you power the pedal on. If the USB cable is the only power it gets, plug the cable in while you hold the switch. The display shows `MC10 CONFIG MODE` / `OPEN THE EDITOR`. The pedal does not play in this mode — it only waits for the editor.
3. Open the editor in **Chrome or Edge** and click **Connect pedal** 1. Pick the port that appears. The display changes to `EDITOR CONNECTED`, and the editor tells you the firmware version and the name of the config the pedal is using now. The list **Configs on the pedal** fills in.
4. **Choose a starting point. Read from pedal** 2 loads what the pedal is using now. Otherwise you start from the layout the firmware comes with, or open a file you saved earlier with **Open a config file**.
5. **Make your changes** in the editor. The Checks at the bottom of the page must show no red items — until then **Send to pedal** stays switched off.
6. Click **Send to pedal** 3 and give the config a name — `THE BAND`, `WEDDING GIG`. This is the name you will see on the pedal. A name that already exists is replaced (the editor asks first); the pedal holds ten configs at most.
7. **The config is saved, but the pedal does not switch to it yet.** The display shows `SAVED TO LIST`. To use it, click **Use this** on its line in **Configs on the pedal** 4, then press **Restart pedal** 5. The pedal restarts with that config.
8. **Check.** When the pedal has restarted, the info row shows the name of the running config in blue, just under the effect grid. If it says `FACTORY DEFAULT`, the stored config was missing or damaged and the pedal runs its built-in settings — send yours again.

> CONFIG MODE ENDS WITH A RESTART
>
> The pedal does not go back to playing by itself. Press **Restart pedal** in the editor, or switch the pedal off and on. If a transfer fails, the display says `CONFIG ERROR` and nothing on the pedal has changed — just send it again.

Choosing a config at start-up — no computer needed

Different band tonight, no laptop in the case. The pedal switches to the config you pick; nothing can be changed or deleted this way.

1. **Power the pedal on normally** — do not hold any switch.
2. While the **firmware version** is on the display (about three seconds), **turn the encoder knob**. The list of configs appears, starting on the one that is running.
3. **Turn the knob** to step through the list. The last item is `EXIT`.
4. **Short press** on a config: it becomes the pedal's config and the pedal restarts with it. Short press on `EXIT`: the pedal carries on with the one that was already running.

If you do not touch the knob, the pedal starts exactly as before. Once the list is open it waits for you — it will not time out in the middle of a show.

No Chrome or Edge? Use the drive

The editor opens in any browser and you can build a layout in it. Only the buttons that talk to the pedal — **Connect pedal**, **Send**, **Read** — need Chrome or Edge on a computer. Safari, Firefox and tablets cannot do that. In that case the editor only makes the file, and you carry it across the way you would copy a file to a USB stick.

1. In the editor click **Save config for pedal**. The browser saves a file called `MC10CFG.TXT` in your **Downloads** folder.
2. Connect the pedal with the USB cable and hold **SWITCH 2** while you power it on. The display shows `MC10 USB DRIVE`, and the pedal appears on the computer as an ordinary drive.
3. In the file manager (Explorer, Finder) copy `MC10CFG.TXT` from Downloads onto that drive.
4. **Eject** the drive the way you eject a USB stick. The pedal checks its config and restarts.
5. **Pick the file.** If the pedal had no config before, it uses this one straight away. If it already holds configs, the copied file only joins the list and the pedal keeps running the one it had: choose the new one at start-up with the encoder (see above).

Keep the saved file: it is also your backup. The editor opens it again with **Open a config file**.

Several configs — one per band, one per venue

The pedal holds up to **ten** configurations. Give each one a name you will recognise — `THE BAND`, `WEDDING GIG`, `HOME` — and switch between them in seconds. Sending a config does not change which one the pedal uses: the new one simply joins the list.

In the editor, **Configs on the pedal** lists what is stored. Every line offers five things:

BUTTON	WHAT IT DOES
Use this	Picks it as the config the pedal starts with, and loads it here for editing
Edit	Loads it for editing without changing what the pedal starts with
Rename	Gives it a new name — the file is renamed with it
Overwrite	Replaces it with what is in the editor now (the editor asks first)
Delete	Removes it from the pedal

Picking or sending does not switch the pedal over on the spot: press **Restart pedal** when you are done, and it comes back with the config you chose.

> THE NAME YOU TYPE BECOMES THE FILE NAME
>
> The pedal stores each config as a file, and file names have to be short and plain: `Élő próba 2` becomes `ELOPROBA.TXT`. The editor shows you the file name before it sends. Full names, accents and all, still appear on the display — they live inside the file.
> Send a config under a name that is already there and it replaces that one. When all ten slots are full, delete one first — the editor tells you before it sends.

The checks at the bottom of the page

The editor checks the layout while you work, and sorts what it finds into two lists.

WHAT IT SAYS	WHAT IT MEANS
Will not work	Something would genuinely misbehave on the FM3 — for example a CC number the pedal itself uses. The switch concerned is marked in red, and **Send to pedal** and **Save config for pedal** stay switched off until you fix it.
Worth a look	It works, but it may not be what you meant — two switches jumping to the same scene, or a label too wide for the display. Nothing is blocked.

A few things the editor simply won't let you set, because they cannot work: switches 1–3 are the effect switches, so only an on/off toggle belongs there; and if a switch has a long press, its short press can only fire when you lift your foot — until then there is no way to tell which one it will be.

Going back to the built-in settings

In config mode, open **Configs on the pedal** and delete the config that is in use. The pedal then starts with its built-in settings and the info row shows `FACTORY DEFAULT`. Keep a copy of your configs with **Save config for pedal** and you can always put them back — the editor opens those files too.

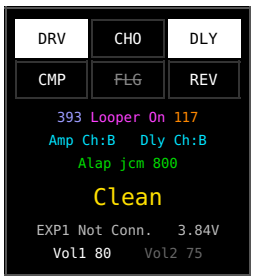
A few things worth knowing

The switch label and the FM3 block are two separate things. The label is just text on the display. Which block a switch actually turns on and off is chosen from the list next to it. If you put a Pitch block on switch 1, change both: the label to `PITCH`, and the block to Pitch. The grid label (`DRV`, `CHO` …) is text as well — it is under Display rows.

One CC number is one function. On the FM3 a CC number does one job, so two different things must not share one. The editor checks this and tells you before you send. The only exception is the six effect slots: they may share a CC, so one switch can turn two blocks on together. **CC 0 is not allowed anywhere** — it is Bank Select, and the pedal itself uses it when it jumps to a preset.

Channel switching can be turned off. If you tend to step on two switches by accident, set the footswitch pairs to off. Then stepping on both does nothing at all — no channel change and no scene change. Each switch still works on its own, and the channel is still shown on the display.

7 Reading the display



effect grid	filled = on · crossed out = not in this preset
info row 1	preset number · looper state · tempo
info row 2	channel switching — and at start-up, the name of the configuration that is running
preset name	from the FM3
the large line	scene name
bottom rows	expression pedals and battery, then Output 1 / Output 2 volume

The large line normally shows the scene name. Switch an effect and it briefly shows that instead — `DRIVE ON` — then returns. Tap the tempo and it shows the BPM the same way.

> BOXES WITH A RED X
> Those blocks aren't in the current preset. Press one and its LED flashes red: there is nothing to switch. This is information, not a fault.

Expression pedals

Plug a pedal in and the controller starts calibrating on its own — nothing to press. Sweep the pedal fully up and down six times. The label blinks while it learns, and stops when it's done. If nothing is plugged in, the controller stops checking that socket.

> WHY SIX SWEEPS
> One movement isn't enough to tell a real pedal from a bit of electrical noise. Six full travels prove it's a pedal and give the controller the true end points.

The encoder — volume and brightness

DO THIS	WHAT HAPPENS
Turn it	Changes the active output volume — CC 16 is Output 1, CC 17 is Output 2
Short press	Switches between Vol1 and Vol2. The active one is shown bright, the other dim
Long press	Enters brightness mode — hold it again to leave

In brightness mode the large line shows `Brightness LEDs`. Turn the encoder to change it, or press briefly to switch to `Brightness Screen`. A long press returns you to the volumes.

> ASSIGN THEM ON THE FM3
> CC 16 and CC 17 do nothing until the FM3 knows about them. Under *Setup* → *MIDI/Remote* → *External*, point them at Output 1 Volume and Output 2 Volume. Both numbers can be changed in the editor.
>
> Handy on stage: Output 1 to front of house, Output 2 to your monitor — two independent levels under one knob.

8 Switch layout

Why it is laid out like this

This layout is a choice, not a limitation. It is the author's concept of how a foot controller should work live: the preset and the scene set the sound, and during the song the **effects are switched on and off by foot, in stomp-box style**, as if six little stompboxes sat at your feet. Each effect has its own place and its own LED, so you never have to think about what a press will do.

Short press, long press. Switches 1–3 hold two effects each: a short press is one, a long press the other. That is how three switches make six effects. A switch with a long press fires its short press when you lift your foot, because until then there is no way to tell which one you meant.

Two switches together. Neighbouring pairs do the things you rarely need but want at hand: Amp and Delay channel A–D, the looper block on and off, and entering the looper. You step on two switches at once, so the single presses stay free for playing. If you tend to hit two by accident, the channel pairs can be turned off in the editor.

The looper, live. Step on switches 1 and 6 together and the pedal turns into a looper: REC, PLAY and STOP sit on three neighbouring switches (7, 8, 9) in the bottom row, ONCE on 10, and REVERSE, HALF and ERASE on the top row (2, 3, 4). Step on 1 and 6 again and you are back to the effects. Nothing needs to be set up on stage, and the looper layout is fixed, so it is always the same under your foot.

> THE LOOPER FOLLOWS THE FM3 (FIRMWARE 6.21 AND LATER)
> In looper mode the pedal keeps asking the FM3 what its looper is doing. Whether you press a button on the pedal or on the FM3 itself, the LEDs and the display show the same: REC, PLAY, OVERDUB (REC and PLAY lit together), REVERSE and HALF. Switch 10 lights up while ONCE plays the loop and goes dark by itself when it ends. Turning the looper block on or off on the FM3 shows exactly as if you had stepped on UP + DOWN. If you change preset on the FM3 while in looper mode, the pedal leaves looper mode and loads the new preset. Without a MIDI connection to the FM3 (NOT SYNC) the LEDs only show your own presses.

What the firmware comes with

All of it can be changed in the editor — see section 6. Switches are numbered 1–5 on the top row and 6–10 on the bottom row; the table also uses the labels UP (5), A–D (6–9) and DOWN (10).

SWITCH	SHORT PRESS	LONG PRESS
1	Drive	Compressor
2	Chorus	Flanger
3	Delay	Reverb
4	Tap tempo	—
UP	Preset +	—
A	Scene 1	Scene 5
B	Scene 2	Scene 6
C	Scene 3	Scene 7
D	Scene 4	Scene 8
DOWN	Preset −	—

> ONLY SWITCHES 1–3 DRIVE THE EFFECT GRID
> Their short and long press are the six grid boxes. The other switches have fixed roles. If you want a completely free layout, PaintAudio's own firmware (Geek / Super mode) is the tool for that.

Two switches together

COMBINATION	WHAT IT DOES
UP + DOWN	Looper block bypass
1+A	Enter or leave looper mode
A+B	Amp channel A ↔ B
B+C	Amp channel C ↔ D
C+D	Delay channel A ↔ B
D + DOWN	Delay channel C ↔ D

The four channel combinations can be switched off in the editor — see section 6.

> THE ENCODER AT START-UP
> While the firmware version is on screen, turning the encoder opens the list of configs stored on the pedal (section 6). At any other time the encoder does what section 7 describes: volume, and brightness on a long press.

9 Troubleshooting

The display says `NOT SYNC`

The controller can't reach the FM3. Check in this order: (1) are **both** MIDI cables in, pedal OUT → FM3 IN and FM3 OUT → pedal IN? This is the most common one. (2) is the FM3 switched on and finished booting? (3) does the MIDI channel match? Then tap the TAP switch (4) once: if the FM3 answers, the pedal syncs again and nothing is sent to the FM3 (firmware 6.21.5 and later). A preset switch retries too.

Presets don't change

Program Change is switched off on the FM3. *Setup → MIDI/Remote → PC Configuration.*

An effect switch does nothing and its LED flashes red

That block isn't in the current preset. The controller is telling you there's nothing to switch. Try a preset that contains the block. If it happens with every preset, check in the editor that the switch is pointed at the right FM3 block.

The editor doesn't find the pedal

Four things to check: the pedal must be in config mode (**hold switch 1 while powering on** — the display says `OPEN THE EDITOR`); the browser must be Chrome or Edge on a computer; the USB cable must carry data, not just charge; and no other program may be holding the port — close any serial monitor. If it still won't connect, use the file route with switch 2 (section 6).

The config list doesn't appear in the editor

The editor writes what went wrong next to the buttons — read that line first. The usual causes: the pedal isn't in config mode, the browser isn't Chrome or Edge on a computer, or the pedal's firmware is older than 6.19 and only holds a single config.

The pedal says `CONFIG ERROR`

The settings arrived damaged, so the pedal discarded them and kept what it had. Nothing is half-applied. Send them again.

The display says `NO FS`

The firmware was built without a storage partition, so there is nowhere to keep a configuration. The pedal still plays normally with its built-in settings. This only happens with a self-built firmware.

The pedal starts with the wrong config

Sending a config doesn't make it the active one. Pick it in **Configs on pedal** and press **Restart pedal**, or turn the encoder while the version is on screen at start-up. If the display shows `FACTORY DEFAULT`, the stored config is missing or damaged and the pedal is running its built-in settings — send yours again.

Everything feels slow after changing preset

Normal on heavy presets: the FM3 is busy loading and can't answer until it's finished. If the FM3's own screen flashes CPU LIMIT, that preset is overloaded — that's the FM3, not the controller.

The pedal won't appear as a drive

Try a different USB cable first — many are charge-only and carry no data at all. For the pedal's own drive, hold **switch 2** while powering on. For bootloader mode, use the browser tool from the official update document — Chrome only, Safari will not work.

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